

GCSE subject knowledge quiz 2025

This is the answer sheet for the subject knowledge quiz. For each question answer using the multiple choice format clicking on the option you think is correct. Please rank your confidence /10 that your answer is correct. 10= extremely confident 1 = I am guessing. Remember the premise is not the score you get but to identify specific areas of strength and target areas for development. Please answer each question and honestly rank your confidence.

When you submit this form, it will not automatically collect your details like name and email address unless you provide it yourself.

Required

1

please enter your first name and surname

(1 Point)

2

Please enter your St Mary's Email address

3

What is a light microscope?

(1 Point)

The fineness of detail that can be seen in an image. The higher this is, the more detail the image holds

A microscopic infectious non cellular pathogen

A scientific device used to magnify images with two lenses- the objective lens and the eyepiece

A glass slide with a scale etched on it. It is used to calibrate the eyepiece of a microscope

A device that uses visible light and a series of lenses to produce an enlarged image of an object.

4

Please rank your confidence that the answer you have given is correct.

5

Below shows the method for producing monoclonal antibodies Stimulate mouse lymphocytes to make an antibody Create hybridoma cells to form the lymphocytes Hybridoma cells make the antibody Antibody is then collected and purified Why are hybridoma cells made rather than just using lymphocytes?

(1 Point)

Hybridoma cells make only one specific antibody

Lymphocytes on their own will only produce the antigen.

Hybridoma cells make antibodies release them into solution

It is much easier to collect and purify antibody from hybridoma cells

Hybridoma cells are made from tumour cells which divide more rapidly so more antibody is made

6

Please rank your confidence that the answer you have given is correct.

7

Which statement is true about mitosis?

(1 Point)

The two daughter cells are produced, each has the same number of chromosomes as the parent cells

The two daughter cells are produced, each has half the number of chromosomes as the parent cells.

Four daughter cells are produced, each has the same number of chromosomes as the parent cells.

The two daughter cells are produced, each has the sum of all the chromosomes from each of the parent cells.

8

Please rank your confidence that the answer you have given is correct.

9

What happens at the stage that is before mitosis in the cell cycle?

(1 Point)

The cytoplasm separates

A period of growth during which time DNA is checked for errors

A period of DNA synthesis

A period of RNA synthesis

10

Please rank your confidence that the answer you have given is correct.

11

Which of these is a communicable disease?

(1 Point)

HIV

Diabetes

Colorectal cancer

none of the above

12

Please rank your confidence that the answer you have given is correct.

13

What is a neurological disorder?

(1 Point)

A condition affecting the nervous system- the brain, spinal cord or nerves and may also affected an associated system

. A chronic neurodegenerative disease which worsens over time.

A condition with affects mental health and causes people to experience symptoms such as depression

A condition where the supply of blood to the heart is suddenly stopped.

14

Please rank your confidence that the answer you have given is correct.

15

Which of the following would not lead to an earlier crop of tomatoes?

(1 Point)

Increasing the heating during the day.

Increasing the carbon dioxide at night

increasing the light intensity at night

increasing the carbon dioxide concentration during the day.

16

Please rank your confidence that the answer you have given is correct.

17

Which plants make food

frutis and vegetables only

all plants

only plants that are eaaten by other animals

only plants found on land

18

Please rank your confidence out of 10 that your answer is correct.

19

Environmental change can be measured using living indicators. Which example is not a living indicator?

(1 Point)

Lichen

Mayfly larvae

Sludgeworms

Amount of rainfall.

20

Please rank your confidence that the answer you have given is correct.

21

What are three examples of conditions controlled by homeostasis?

(1 Point)

body temperature, nervous and chemical responses.

Blood glucose concentration, body temperature and water levels.

Blood glucose concentration, body temperature and nervous responses.

Blood glucose concentration, nervous and chemical responses.

22

Please rank your confidence that the answer you have given is correct.

23

Define the term phenotype.

(1 Point)

The location in the DNA occupied by a particular gene.

The visible characteristics of an organism which occur as a result of its genes.

A self-regulating process in development, such that the organism grows up to have much of the same form independent of external influence.

An individual having two different alleles at a genetic locus.

Members of the family Hominidae, which includes only modern humans and ancestors since the human lineage split from the apes.

24

Please rank your confidence that the answer you have given is correct.

25

How would you rank your confidence for your answers in the biology section?

26

Below are some statements about the particle model for the gas state. All the statements are correct.

A Most of the particles are far from each other. **B** The particles move around rapidly in all directions. **C** The particles collide with the walls of the container they are in. **D** Most of the particles are too far apart to exert any force on each other **E** If you heat a gas, the average speed of the particles gets bigger.

Which of the statements above help to explain that Substances in the gas state are fairly easy to compress.

(1 Point)

A

B

C

D

E

27

Please rank your confidence that the answer you have given is correct.

28

Below are some statements about the particle model for the gas state. All the statements are correct.

A Most of the particles are far from each other. **B** The particles move around rapidly in all directions. **C** The particles collide with the walls of the container they are in. **D** Most of the particles are too far apart to exert any force on each other **E** If you heat a gas, the average speed of the particles gets bigger.

Which of TWO the statements above help to explain that Substances in the gas state spread out to fill the whole space they are in.

(1 Point)

A

B

C

D

29

Please rank your confidence that the answer you have given is correct.

30

Which of the following solutions is likely to have the highest pH?

(1 Point)

A weak dilute acid.

A strong dilute acid.

A weak concentrated acid.

A strong concentrated acid.

31

Please rank your confidence that the answer you have given is correct.

32

2.3g of sodium burns completely in oxygen. What is the maximum theoretical mass of sodium oxide that can be produced.

(1 Point)

2.3g

3.9g

3.1g

6.2g

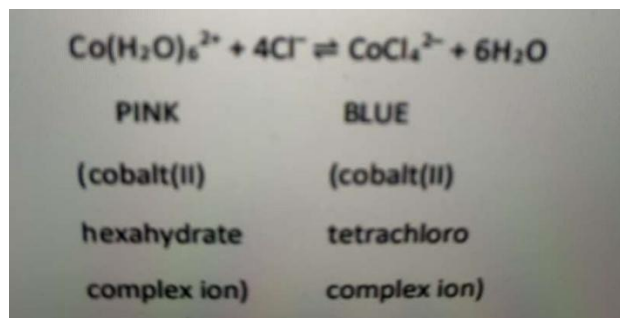
33

Please rank your confidence that the answer you have given is correct.

34

The picture opposite shows an example of a reversible reaction. In this example, what would happen to the colour if more concentrated hydrochloric acid is added?

(1 Point)



A pink colour is produced

A blue colour is produced

A purple colour is produced

it will not change the colour.

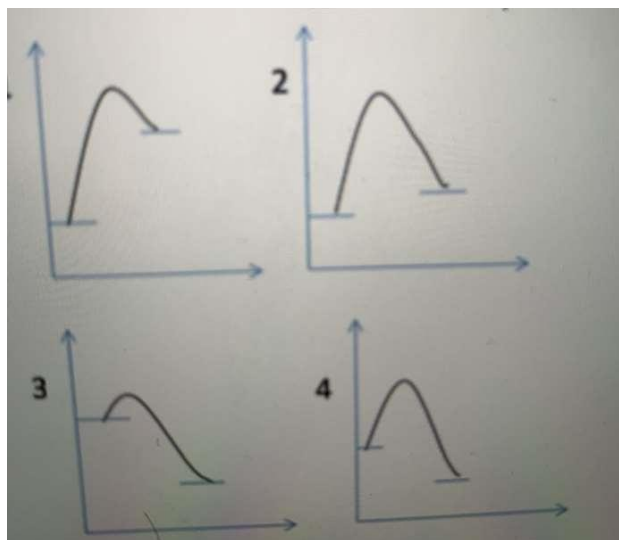
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Please rank your confidence that the answer you have given is correct.

36

Refer to the graphs opposite. Which reaction profile shows an exothermic reaction with a large activation energy? Y axis is energy x axis is time/ progress of the reaction.

(1 Point)



1

2

3

4

37

Please rank your confidence that the answer you have given is correct.

38

The reaction between hydrogen and oxygen is exothermic. In what process is energy released?

(1 Point)

breaking down hydrogen molecules only

breaking down oxygen molecules only

making water molecules

breaking both hydrogen molecules and oxygen molecules

39

Please rank your confidence that the answer you have given is correct.

40

A car has 60kg of fuel in its tank. After driving the tank is empty. What is the mass of exhaust gases produced during driving?

(1 Point)

the same as the mass of the starting fuel

As the fuel reacts with oxygen, more than the starting mass

As the products are a gas, less than the starting mass of fuel

As some of the fuel is converted to energy the mass of gaseous products is less than the starting fuel.

41

Please rank your confidence that the answer you have given is correct.

42

what does a negative enthalpy value mean?

(1 Point)

Overall energy is absorbed from the surroundings during the reaction

Overall energy is released to the surroundings

the temperature of the reaction decreases

More bonds are made than broken

43

Please rank your confidence that the answer you have given is correct.

44

Please rank your confidence that the answer you have given is correct.

45

Which list only contains empirical formulae? Assume all numbers are subscript.

(1 Point)

C₃H₆, I₄O₁₀, Al₂Cl₆

CH₂, P₂O₅, MgO

C₃H₄, C₂H₆, C₁₂H₂₂O₁₁

P₄O₁₀, Al₂Cl₆, MgO

46

Please rank your confidence that the answer you have given is correct.

47

Why is ice hard, but liquid water is runny?

(1 Point)

Water particles can't move in ice, making it hard.

Ice is colder than liquid water.

Chemical bonds in ice are stronger.

The strongest forces between molecules of water are not broken in ice.

48

Please rank your confidence that the answer you have given is correct.

49

Which of the following is correct?

(1 Point)

The atmosphere is created by plants which photosynthesised

The atmosphere is getting larger due to us burning fossil fuels and releasing extra gasses

The atmosphere formed from gases released by volcanoes

The early atmosphere had a lot more oxygen than today.

50

Please rank your confidence that the answer you have given is correct.

51

Please rank your confidence in your answers on the chemistry section.

52

A wood stove has an efficiency of 0.4. The useful energy output is 104J. How much energy was input?

(1 Point)

41.6J

104.4J

260J

103.6J

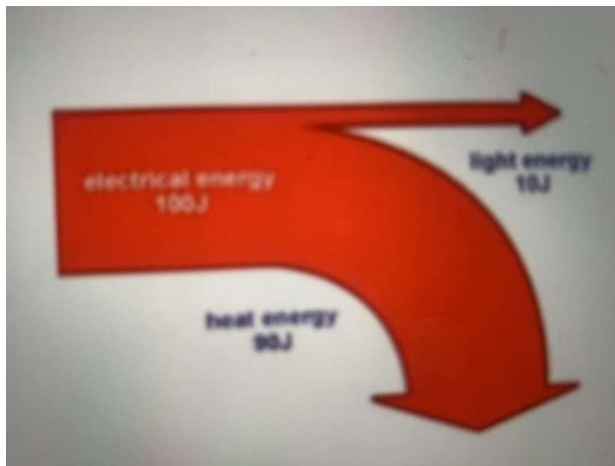
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Please rank your confidence that the answer you have given is correct.

54

Using the Sankey diagram opposite which has an internal input of 100J an output of 10J light and 90J heat, what is the efficiency of the bulb?

(1 Point)



0.1

0.9

1

10

55

Please rank your confidence that the answer you have given is correct.

56

The *change* of energy in a thermal store can be calculated using the equation in the picture:

Imagine you have a 2 kg block of gold that is heated from 20 degrees Celsius to 40 degrees Celsius. The specific heat capacity of the gold block is 130J/Kg/Degrees celcius. What does this calculation tell us about energy in the thermal store of the gold block?

(1 Point)

$$\text{Change in energy} = \text{Mass} \times \text{Specific heat capacity} \times \text{Change in temperature}$$

$$\Delta E = m \times c \times \Delta \Theta$$

It has 5200 J of energy in its thermal store.

It has 10 400 J of energy in its thermal store.

It has an extra 5200 J of energy in its thermal store.

57

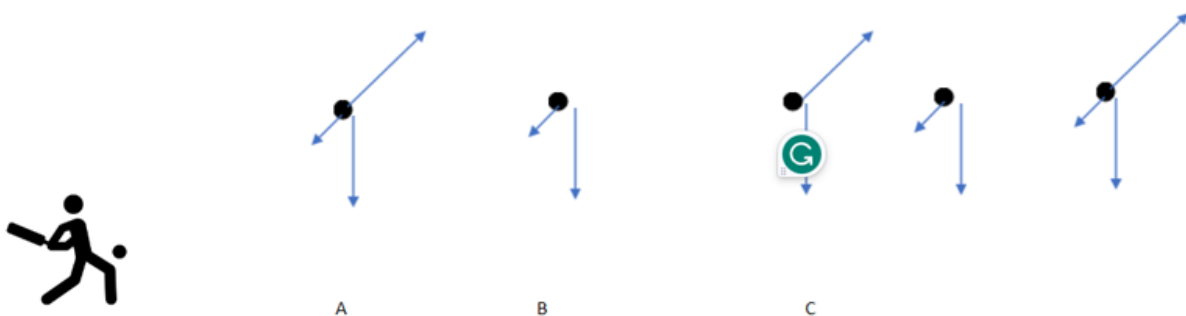
Please rank your confidence that the answer you have given is correct.

58

Refer to the photos below. A ball is hit with a bat and is now travelling diagonally upwards. Which force diagram best represents the forces acting on the ball while it is travelling upwards?

(The length of the arrow indicates the size of each force.)

(1 Point)



A

B

C

D

E

59

Please rank your confidence that the answer you have given is correct.

60

Pam takes two identical sheets of paper. She crumples one into a tight ball. Then she drops the paper ball and the sheet of paper at the same moment from the same height above the ground. Which of the following best explains how the two objects fall? Source: EPSE (Evidence-based Practice in Science Education) project, Robin Millar. © University of York

(1 Point)

There is an air resistance force on the sheet of paper, but not on the paper ball.

The acceleration due to gravity (g) is a constant – for all objects

The air resistance force on the sheet of paper becomes equal to the gravity force at a slower speed than for the paper ball.

The force of gravity on the paper ball is bigger than the force of gravity on the sheet of paper.

The air resistance force on a large flat object is always bigger than on a small compact object.

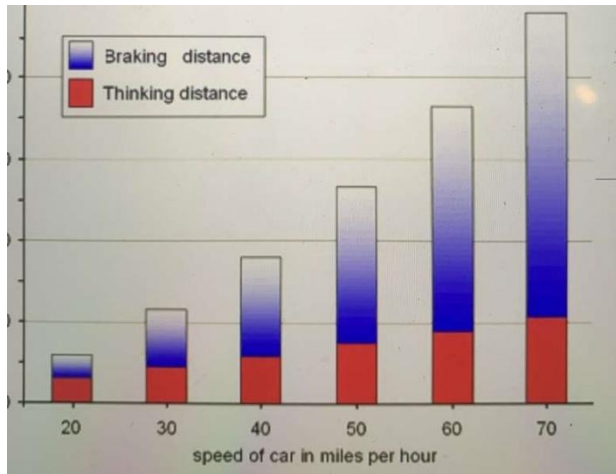
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Please rank your confidence that the answer you have given is correct.

62

Refer to the graph opposite. The following graph shows approximate stopping distances for a typical car, plotted against the speed of the car before braking. What is the main reason the braking distance increases with speed? Source: IOP / BBC Bitesize GCSE Revision

(1 Point)



For higher initial speeds, there is less friction between the tyres and the ground.

For higher initial speeds, there is less friction between the brake pads and the wheels.

For higher initial speeds, there is more friction between the brake pads and the wheels.

For higher initial speeds, the same deceleration takes longer to bring the car to a stop.

For higher initial speeds, there is more friction between the tyres and the ground.

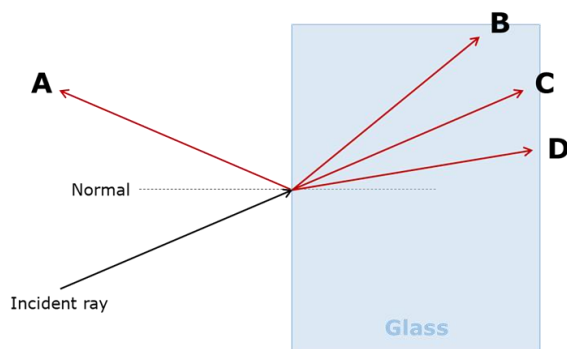
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Please rank your confidence that the answer you have given is correct.

64

Refer to the diagram opposite. Which ray of light shows what happens at a boundary between air and glass?

(1 Point)



A

B

C

D

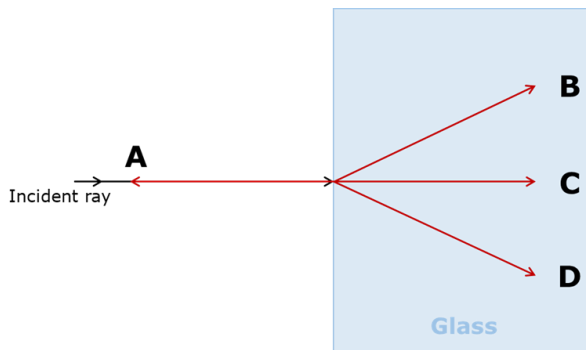
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Please rank your confidence that the answer you have given is correct.

66

Look at the diagram opposite Which ray of light shows what happens at a boundary between air and glass in this context?

(1 Point)



A

B

C

D

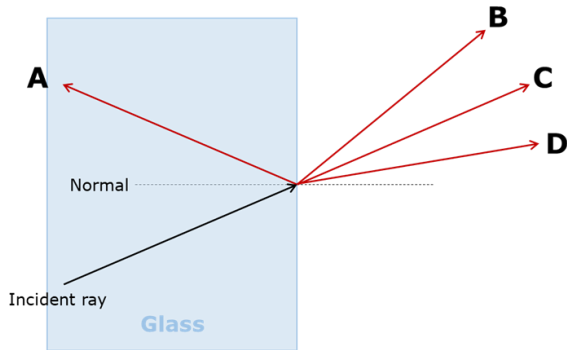
67

Please rate how confident you are that your answer is correct

68

Look at the picture opposite Which ray of light shows what happens at a boundary between glass and air?

(1 Point)



A

B

C

D

69

Please rank how confident you are that your answer is correct

70

A plastic duck in a bath is pushed up and down 3 times in 4 seconds. The crests of the waves it makes are 4 cm apart. What is the wavelength and frequency of the wave produced?

(1 Point)

Wavelength 8cm frequency $\frac{4}{3}$ Hz

Wavelength 8cm frequency $\frac{3}{4}$ Hz

Wavelength 4cm Frequency $\frac{4}{3}$ Hz

Wavelength 4cm Frequency $\frac{3}{4}$ Hz

Wavelength 2cm Frequency $\frac{4}{3}$ Hz

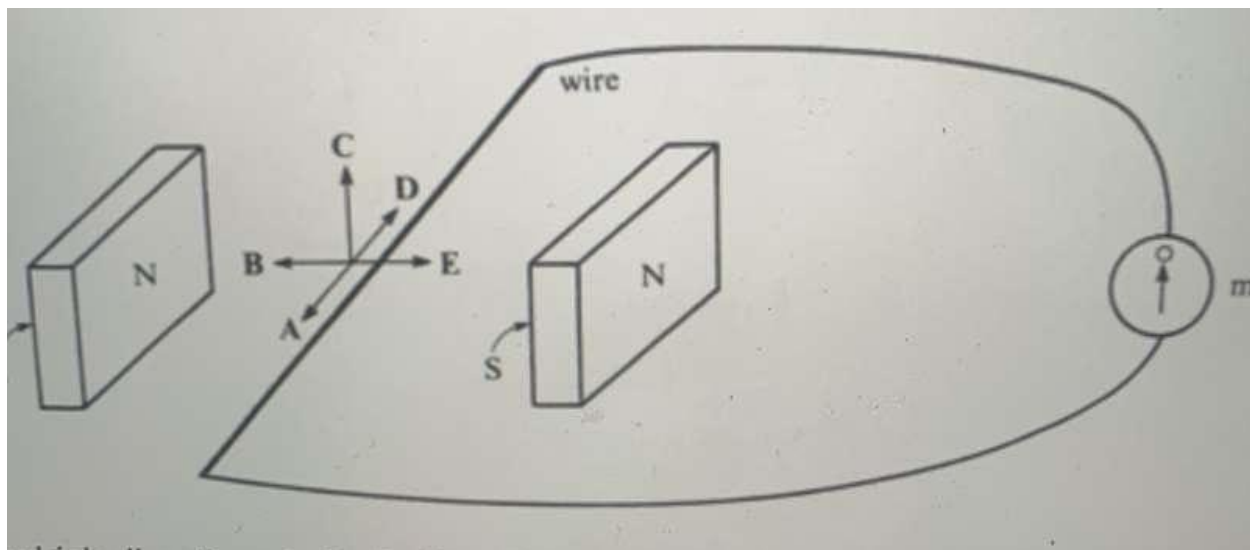
71

Please rank your confidence that the answer you have given is correct.

72

The diagram shows a wire in a magnetic field. It is connected to a voltmeter. In which direction, A, B, C, D or E, should the wire be moved to deflect the pointer of the voltmeter?

(1 Point)



A

B

C

D

E

73

Please rank your confidence that the answer you have given is correct.

74

A radioactive substance emits beta particles. The count rate is measured at one minute intervals and the results are shown in the table below:

Time in seconds	0	60	120
Counts per second	169	60	42

What is the approximate value of the half-life? Source: Nuffield O Level Exam Question

(1 Point)

30s

40s

20s

80s

60s

75

Please rank your confidence that the answer you have given is correct.

76

Please rank your confidence in the physics questions.